Some computations *On maximizing the number of heads...*

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Rewriting the basic recursion

Let us rewrite (1) in the manuscript by using $w_{n,p} := n - v_{n,p}$, i.e. $w_{n,p}$ is the expected number of tails set aside during the experiment. This gives

$$\begin{aligned} w_{n,p} &= n(p^n + (1-p)^n) - np^n - v_{n-1,p}(1-p)^n + \sum_{j=1}^{n-1} \binom{n}{j} p^j (1-p)^{n-j} \Big(\underbrace{n - \max_{1\le i\le j}(v_{n-i,p} + i)}_{=\min_{1\le i\le j} w_{n-i,p}} \Big) \\ &= (w_{n-1,p}+1)(1-p)^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j \Big(\min_{j\le i\le n-1} w_{i,p} \Big). \end{aligned}$$

At least for me, this is a bit easier to grasp than (1) in the manuscript.

Some notes on strategy A

Recall that strategy A consists of only setting aside one coin, unless all are heads. In the latter case, set aside all coins. (Also, recall that strategy A is optimal iff $v_{n,p} \ge v_{n-1,p} + 1$ for all n.) This is the case iff

$$w_{n,p} = n - v_{n,p} \le n - v_{n-1,p} - 1 = w_{n-1,p},$$

i.e. the map $i \mapsto w_{i,p}$ is decreasing, and we find $\min_{j\le i\le n-1} w_{i,p} = w_{n-1,p}$.) For strategy A, we have

$$w_{n,p} = (1-p)^n + (1-p^n) w_{n-1,p}$$

and this recursion is readily solved by

$$w_{n,p} = \sum_{k=1}^{n} (1-p)^k (1-p^{k+1}) \cdots (1-p^n).$$

In fact, this formula is not surprising. We can write

$$\begin{aligned} w_{n,p} &= \mathbb{E}\Big[\sum_{k=1}^{n} 1_{\text{set aside a tail when there are } k \text{ coins left}} \Big] \\ &= \sum_{k=1}^{n} \mathbb{P}(\text{set aside a tail when there are } k \text{ coins left}), \end{aligned}$$

and we set aside a tail when there are k coins left iff there were less than ℓ heads when having $\ell = k+1, ..., n$ coinds left (otherwise we would not have come to the situation of having k coins left), and there are all tails when throwing k coins.

A martingale

Even a bit more can be said for strategy A: Let $X_1, X_2, ...$ be the (random) total number of heads which are set aside when throwing $1, 2, ...$ coins and following strategy A. If we throw n coins and $1 \leq X_n < n$, then $X_n = 1 + X_{n-1}$ etc. We use that, for $\mathcal{F}_{n-1} = \sigma(X_1, ..., X_{n-1})$, and with $Z_n := n - X_n$, with a similar calculation as above,

$$\begin{aligned}\mathbb{E}[Z_n|\mathcal{F}_{n-1}] &= (Z_{n-1}+1)(1-p)^n + (1-p^n-(1-p)^n)Z_{n-1} \\ &= (1-p)^n + (1-p^n)Z_{n-1}.\end{aligned}$$

In this situation, we can use the following martingale[1]:

Lemma 0.1. *Let $Z_0 = 0, Z_1, ...$ and $\mathcal{F}_{n-1} := \sigma(Z_1, ..., Z_n)$. If $\mathbb{E}[Z_n|\mathcal{F}_{n-1}] = a_n + b_n X_{n-1}$ for all $n = 1, 2, ...$, then with $c_n = b_n c_{n-1} + a_n$,*

$$\frac{Z_n - c_n}{b_1 \cdots b_n} \text{ is a martingale.}$$

Note that the recursion for c_n is solved by

$$c_0 = 0 \text{ and } c_n = \sum_{k=1}^{n} b_n \cdots b_{k+1} a_k.$$

In particular,

$$\mathbb{E}[Z_n] = b_1 \cdots b_n \frac{\mathbb{E}[Z_n] - c_n}{b_1 \cdots b_n} + c_n = b_1 \cdots b_n(\mathbb{E}[Z_0] - c_0) + c_n = c_n = \sum_{k=1}^{n} b_n \cdots b_{k+1} a_k.$$

Proof. We write directly

$$\mathbb{E}\Big[\frac{Z_n - c_n}{b_1 \cdots b_n}\Big|\mathcal{F}_{n-1}\Big] = \frac{a_n + b_n X_{n-1} - b_n c_{n-1} - a_n}{b_1 \cdots b_n} = \frac{X_{n-1} - c_{n-1}}{b_1 \cdots b_{n-1}},$$

which proves the martingale property. For the recursion formula, we note that

$$c_n = b_n\Big(\sum_{k=1}^{n-1} b_{n-1} \cdots b_{k+1} a_k\Big) + a_n,$$

and we are done. □

So, we have that

$$\frac{Z_n - \sum_{k=1}^{n}(1-p^n)\cdots(1-p^{k+1})(1-p)^k}{(1-p)\cdots(1-p^n)}$$

is a martingale. Since the denominator is bounded away from 0 for all p, and the sum in the numerator converges as $n \to \infty$, we find that the martingale is bounded from below, and hence converges almost surely.

[1] Recall that $(Y_n)_{n=1,2,...}$ is a martingale iff $\mathbb{E}[Y_n|\mathcal{F}_{n-1}] = Y_{n-1}$.

Strategy C: Always set aside as many heads as possible: Starting with n coins and with $k \leq n$, let $p_n(k)$ be the probability that there are k coins left at some step. Then, we get the recursion

$$p_n(k) = p_n(k+1)(1-p)^{k+1} + \sum_{\ell=k+1}^{n} p_n(\ell)\binom{\ell}{k}p^k(1-p)^{\ell-k}.$$

Hence, with $\psi_n(s) := \sum_{k=0}^{\infty} p_n(k)s^k$ and $s \in (0,1)$,

$$\begin{aligned}\psi_n(s) = \sum_{k=0}^{\infty} p_n(k)s^k &= \frac{1}{s}\sum_{k=0}^{\infty} p_n(k+1)(s(1-p))^{k+1} + \sum_{\ell=1}^{n} p_n(\ell)\sum_{k=0}^{\ell-1}\binom{\ell}{k}p^k(1-p)^{\ell-k}s^k \\ &= \frac{\psi_n(s(1-p))-1}{s} + \psi_n(1-p+ps) - \psi_n(ps),\end{aligned}$$

where we have used that $p_n(0) = 1$ (since the experiment certainly ends with 0 coins).

Let us change the game a bit: you only win if all coins come up heads, otherwise you lose. What is the best strategy for winning (depending on p)?

In this case, we get the Bellman equation for $w_{n,p} = \max_s \mathbb{P}_s(\text{winning with } n \text{ coins left})$, where $\mathbb{P}_s$ is the law when playing strategy s with n coins. We get

$$w_{n,p} = p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j \max_{j \le i \le n-1} w_{i,p} \tag{1}$$

eq:wn

Let us start simple and compute the first few terms. Clearly, $w_{1,p} = p$. Then,

$$\begin{aligned} w_{2,p} &= p^2 + 2p^2(1-p) = p^2(3-2p), \\ w_{3,p} &= p^3 + 3p(1-p)((1-p)w_{2,p} + p(w_{1,p} \vee w_{2,p})). \end{aligned}$$

Here, note that $w_{1,p} > w_{2,p}$ iff $2p^2 - 3p + 1 = (1-2p)(1-p) > 0$, i.e. $p < 1/2$. So, in this case, if we roll three coins and two come up heads, it is best to set aside both of them, whereas for $p > 1/2$, it is better to set aside only one.

We can show the following:

11 **Lemma 0.2.** *Let $p > 1/2$. Then, for all $n = 2, 3, ...,$*

$$w_{n,p} > w_{n-1,p} \quad \text{and} \quad w_{n-1,p} < \frac{p^n}{(1-p)^n + p^n}.$$

Corollary 0.3. *Let $p > 1/2$. Then $n \mapsto w_{n,p}$ is increasing.*

Proof of Lemma 0.2. First, note that

$$n \mapsto \frac{p^n}{(1-p)^n + p^n} = \frac{1}{1 + ((1-p)/p)^n} \text{ is increasing} \tag{2}$$

eq:in1

since $(1-p)/p < 1$.

We proceed by induction. For $n = 2$, we have already shown $w_{2,p} > w_{1,p}$ as well as $w_{1,p} = p = \frac{p}{(1-p)+p} < \frac{p^2}{(1-p)^2+p^2}$ by (2). Next, assume that the assertion holds for $w_{2,p}, ..., w_{n-1,p}$. Then,

$$\begin{aligned} w_{n,p} &= p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j w_{n-1,p} \\ &= p^n + (1 - p^n - (1-p)^n) w_{n-1,p} = w_{n-1,p} + p^n(1 - w_{n-1,p}) - (1-p)^n w_{n-1,p}. \end{aligned}$$

Clearly, we have $w_{n,p} > w_{n-1,p}$ iff

$$\frac{(1-p)^n}{p^n} < \frac{1 - w_{n-1,p}}{w_{n-1,p}} = \frac{1}{w_{n-1,p}} - 1, \quad \text{i.e.} \quad w_{n-1,p} < \frac{p^n}{(1-p)^n + p^n},$$

which holds by induction hypothesis. Last, we find that

$$\begin{aligned} w_{n,p} &= p^n + (1 - p^n - (1-p)^n) w_{n-1,p} \\ &< p^n \Big(1 + \frac{1 - p^n - (1-p)^n}{(1-p)^n + p^n}\Big) = \frac{p^n}{(1-p)^n + p^n} < \frac{p^{n+1}}{(1-p)^{n+1} + p^{n+1}} \end{aligned}$$

by (2) and we are done.

□

Let us study two concrete strategies:

(A) Always put aside one coin (with head) if there is one, unless all are head. In this case, put aside all (and win).

(B) Always put aside as many coins as there are heads.

The probability for a win with strateggy (A) is

$$p_n = p^n + (1 - (1-p)^n - p^n)p_{n-1},$$
$$p_n = \sum_{k=1}^{n} p^k (1 - (1-p)^{k+1} - p^{k+1}) \cdots (1 - (1-p)^n - p^n).$$

As $n \to \infty$, this gives approximately

$$p_n = \sum_{k=1}^{\infty} p^k \prod_{\ell=k+1}^{\infty} (1 - (1-p)^\ell - p^\ell)$$

For (B), we note the following: Every coin has an independent geo(p)-distribution for the first time it comes uo heads. Let $T_1, T_2, ...$ be these independent random variables. Then, we can describe the probability to win as

$$p_n = \mathbf{P}(\text{there is } t = 1, 2, ... \text{ such that } T_1, ..., T_n \leq t \text{ and for all } s \leq t \text{ there is } i \text{ with } T_i = s).$$

$$\begin{aligned}
p_n &= \sum_{t=1}^{n} \sum_{\substack{k_1,\dots,k_t \geq 1 \\ k_1+\dots+k_t=n}} \binom{n}{k_1 \cdots k_t} \prod_{s=1}^{t} \Big((1-p)^{s-1}p\Big)^{k_s} \\
&= \Big(\frac{p}{1-p}\Big)^n \sum_{t=1}^{n} \sum_{\substack{k_1,\dots,k_t \geq 1 \\ k_1+\dots+k_t=n}} \binom{n}{k_1 \cdots k_t} (1-p)^{\sum_{s=1}^{t} s k_s}
\end{aligned}$$

With $p_0 = 1$,

$$\begin{aligned}
p_n &= \sum_{k=0}^{n-1} \binom{n}{k} (1-p)^k p^{n-k} p_k. \\
\sum_{n=1}^{\infty} p_n s^n &= \sum_{k=0}^{\infty} \sum_{n=k+1}^{\infty} \binom{n}{k} (1-p)^k p^{n-k} p_k s^n \\
&= \sum_{k=0}^{\infty} \frac{1}{k!} ((1-p)s)^k p_k \sum_{n=k+1}^{\infty} n \cdots (n-k+1)(ps)^{n-k} \\
&= \sum_{k=0}^{\infty} \frac{1}{k!} ((1-p)s)^k p_k \frac{d^k}{dx^k} \sum_{n=1}^{\infty} x^n \Big|_{x=ps} \\
&= \sum_{k=0}^{\infty} \frac{1}{k!} ((1-p)s)^k p_k \frac{d^k}{dx^k} \Big(\frac{1}{1-x} - 1\Big)\Big|_{x=ps} \\
&= \frac{ps}{1-ps} + \sum_{k=1}^{\infty} \frac{1}{k!} ((1-p)s)^k p_k (-1)^k k! (1-ps)^{-(k+1)} \\
&= \frac{ps}{1-ps} - \frac{1}{1-ps} \sum_{k=1}^{\infty} p_k \Big(\frac{-(1-p)s}{1-ps}\Big)^k
\end{aligned}$$

Note that

$$\begin{aligned}
&\mathbf{P}(\max_{i=1,\dots,n} T_i \leq t - \log_{1-p} n) = (1 - (1-p)^{t - \log_{1-p} n})^n \approx \exp(-(1-p)^t), \\
&\mathbf{P}(\max_{i=1,\dots,n} T_i = \lfloor t - \log_{1-p} n \rfloor) \approx \exp(-(1-p)^t) - \exp(-(1-p)^{t-1}), \\
&\mathbf{P}\Big(\bigcup_{i=1}^{n} \{T_i = a - \log_{1-p} n\}\Big) = 1 - \prod_{i=1}^{n} \Big(1 - (1-p)^{a - \log_{1-p} n - 1} p\Big) = 1 - \Big(1 - (1-p)^{a-1} p/n\Big)^n \\
&\qquad \approx 1 - \exp\Big(-(1-p)^{a-1} p\Big) \\
&\qquad p_n \approx \sum_{t=-\infty}^{\infty} \big(\exp(-(1-p)^t) - \exp(-(1-p)^{t-1})\big) \prod_{a=-\infty}^{t-1} \Big(1 - \exp\Big(-(1-p)^{a-1} p\Big)\Big)
\end{aligned}$$

$X \sim \exp(1)$, so

$$\mathbf{P}(ts \leq X \leq (t+1)s) = e^{-ts} - e^{-(t+1)s} = e^{-ts}(1 - e^{-s}).$$

For $Z := \max_{i=1,\dots,n} X_i$,

$$\begin{aligned}
\mathbf{P}(Z \in dz) &= \frac{d}{dz}(1-e^{-z})^n = ne^{-z}(1-e^{-z})^{n-1}, \\
\mathbf{P}(Z - \lfloor Z \rfloor < x) &= \sum_{k=0}^{\infty} \int_k^{k+x} ne^{-z}(1-e^{-z})^{n-1} dz = \sum_{k=0}^{\infty} (1-e^{-z})^n \Big|_{z=k}^{z=k+x} \\
&= \sum_{k=0}^{\infty} (1-e^{-(k+x)})^n - (1-e^{-k})^n \\
&= \sum_{k=0}^{\infty} (1+e^{-x})^n (1-e^{-k})^n + -(1-e^{-k})^n
\end{aligned}$$

For large n,

$$\begin{aligned}
p_n &:= \sum_t \mathbf{P}(\text{success}, \max_{i=1,\dots,n} T_i = t) \\
&\approx \sum_t \mathbf{P}(\bigcap_k \bigcup_i 1_{T_i = t-k}, \max_{i=1,\dots,n} T_i = t)
\end{aligned}$$

Heuristics with $p = 1 - e^{-s}$, so $1 - p = e^{-s}$

$$p_\infty \approx \prod_{k=1}^{\infty} (1 - e^{-ks}) = \prod_{k=1}^{\infty} (1 - (1-p)^k)$$

$$\begin{aligned}
&a_1 = 1, \\
&a_2 = p + (1-p)p, \\
&a_3 = p^2 + p^2(1-p) + (1-p)^2 a_2 p_n \qquad \approx p^2 + 2p(1-p)p + (1-p)^2
\end{aligned}$$

Here is another probabilsitic model capturing the model: Take $X_1, \dots, X_n \sim \exp(1)$ independently, and independent Poisson processes $P_1, \dots, P_n$ with rate $\lambda > 0$ with $p = 1/(1+\lambda)$ or $\lambda = (1-p)/p$. Then, by construction, $T_i = |\{t \le X_i : t \in P_i\}| \sim geo(p)$. Rather, let us assume that there is a single Poisson process P, which makes the resulting geometric RVs dependent. Then we want to compute the probability that between any two clicks of P there is at least one i such that X_i runs off. This can be computed via

$$\prod_{k=1}^{n} \frac{k}{k+\lambda}.$$

Let $p_{n,t}$ be the probability of a win when starting with n coins after exactly t coin tosses. Then,

$$p_{n+1,t} = p_{n,t}(1-(1-p)^t) + p_{n,t-1}(1-p)^{t-1}p$$

So, with $a_n(s) = \sum_t p_{n,t} s^t$,

$$a_{n+1}(s) = \sum_t p_{n+1,t} s^t = a_n(s) - a_n(s(1-p)) + spa_n(s(1-p)) = a_n(s) - a_n(s(1-p))(1-sp)$$

$$P(S_{n+1} | \mathcal{F}_n) = 1_{S_n}(1-(1-p)^{T_n+1})$$

$$\mathbf{E}[A_n 1_{S_{n+1}} | \mathcal{F}_n] = A_n 1_{S_n}(1-(1-p)^{T_n+1}) = A_{n-1} 1_{S_n}$$

So, $A_n = \prod_{k=1}^{n} (1-(1-p)^{T_k+1})$

For the case $p < 1/2$ we have the following:

Lemma 0.4. *Let $p < 1/2$. If $w_{n-1,p} = \max_{k=1,\dots,n-1} w_{k,p}$. Then, $w_{n,p} < w_{n-1,p}$.*

Proof. First, note that (since $p < 1/2$),

$$n \mapsto \frac{p^n}{(1-p)^n + p^n} = \frac{1}{1 + ((1-p)/p)^n} \text{ is decreasing,}$$

in particular $\frac{p^n}{(1-p)^n+p^n} < p = w_{1,p}$. From here on, we copy the calculations from Lemma 0.2: We compute

$$\begin{aligned} w_{n,p} &= p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j w_{n-1,p} \\ &= p^n + (1 - p^n - (1-p)^n) w_{n-1,p} = w_{n-1,p} + p^n(1 - w_{n-1,p}) - (1-p)^n w_{n-1,p}. \end{aligned}$$

Clearly, we have $w_{n,p} < w_{n-1,p}$ iff

$$\frac{(1-p)^n}{p^n} > \frac{1 - w_{n-1,p}}{w_{n-1,p}} = \frac{1}{w_{n-1,p}} - 1, \quad \text{i.e.} \quad w_{n-1,p} > \frac{p^n}{(1-p)^n + p^n}.$$

The latter holds since $w_{n-1,p} \geq w_{1,p} = p > \frac{p^n}{(1-p)^n+p^n}$. $\square$

Let us study the case $p < 1/2$. Setting $w_{0,p} := 1$, note that (1) changes to

$$w_{n,p} = \sum_{j=0}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j \max_{j \le i \le n-1} w_{i,p} \tag{3}$$ eq:wn1

$$= p \sum_{j=0}^{n-1} \binom{n-1}{j} p^{n-1-j}(1-p)^j w_j + \sum_{j=0}^{n-2} \binom{n-1}{j} p^{n-1-j}(1-p)^{j+1} w_{j+1} \tag{4}$$

$$= (1-p)^{n-1} p w_{n-1} + \sum_{j=0}^{n-2} \binom{n-1}{j} p^{n-1-j}(1-p)^j((1-p)w_{j+1} + p w_j) \tag{5}$$

Let $p = \frac{1}{2} - \varepsilon$ and $w_n = \frac{1}{2} - \delta_n$. Then,

$$\delta_n = \tfrac{1}{2} - \sum_{j=0}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j (\tfrac{1}{2} - \delta_j) \tag{6}$$ eq:wn1

$$= \tfrac{1}{2}\big(\tfrac{1}{2} + \varepsilon\big)^n - \tfrac{1}{2}(\tfrac{1}{2} - \varepsilon)^n + \sum_{j=1}^{n-1} \binom{n}{j} (\tfrac{1}{2} - \varepsilon)^{n-j} (\tfrac{1}{2} + \varepsilon)^j \delta_j \tag{7}$$

We have

$$\begin{aligned}
&w_{1,p} = p,\\
&w_{2,p} = p^2 + 2p^2(1-p) = p^2(3-2p) = p(3p - 2p^2) < p = w_{1,p},\\
&w_{3,p} = p^3 + 3p(1-p)(p w_{1,p} + (1-p)w_{2,p}) = p^3(1 + 3(1-p)(1 + (1-p)(3-2p)))\\
&\qquad = p^3(1 + 3(1-p)(4 - 5p + 2p^2)) = p^3(13 - 27p - 21p^2 - 6p^3),\\
&w_{3,p} < p^3 + 3p(1-p)p = p^2(3-2p) = w_{2,p},\\
&w_{4,p} = p^4 + 4p^3(1-p)w_{1,p} + 6p^2(1-p)^2 w_{2,p} + 4p(1-p)^3 w_{3,p}
\end{aligned}$$

$$\begin{aligned}
w_{2,p} &= p^2 + 2p(1-p)w_{1,p}\\
w_{3,p} &= p^3 + 3p^2(1-p)w_{1,p} + 3p(1-p)^2 w_{2,p}\\
&\le p^3 + p^2(1-p)w_{1,p} + p(1-p)^2 w_{1,p} + 2p(1-p)w_{1,p}\\
&= (p + p(1-p) + (1-p)^2))p^2 + 2p(1-p)w_{1,p} = w_{2,p}\\
w_{4,p} &= p^4 + 4p^3(1-p)w_{1,p} + 6p^2(1-p)^2 w_{2,p} + 4p(1-p)^3 w_{3,p}\\
&\le p^4 + 4p^3(1-p)w_{1,p} + 3p^2(1-p)^2 w_{1,p} + p(1-p)^3 w_{2,p} + 3p(1-p)^2 w_{2,p}\\
&\le p^4 + p^3(1-p)w_{1,p} + p(1-p)^3 w_{2,p} + 3p^2(1-p)w_{1,p} + 3p(1-p)^2 w_{2,p}\\
&= p^2(p^2 + p^2(1-p) + (1-p)^3) - p(1-p)^3 p(1-2p)(1-p) + 3p^2(1-p)w_{1,p} + 3p(1-p)^2 w_{2,p}\\
w_{4,p} &= p^4 + 4p^3(1-p)w_{1,p} + 6p^2(1-p)^2 w_{2,p} + 4p(1-p)^3 w_{3,p}\\
&\le p^3(p + p(1-p) + (1-p)^2) + 3p^3(1-p)w_{1,p} + 3p^2(1-p)^2 w_{1,p}(p + p(1-p) + (1-p)^2)w_{1,p} + 3p(1-p)^2 w_{2,p}\\
&\le p^3 + 3p^2(1-p)w_{1,p} - 3p^2(1-p)^2 w_{1,p} + 5p^2(1-p)^2 w_{2,p} + 4p(1-p)^3 w_{2,p}\\
&= p^3 + p(1-p)w_{1,p}(3p^2 + 2p(1-p) + (1-p)^2) + 3p(1-p)^2 w_{2,p}
\end{aligned}$$

We have $w_n - w_{n+1} = a_n - b_n$, where $a_0 = 1-p, b_0 = 0$,

$$a_n = (1-p)\sum_{j=0}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j a_j,$$

$$b_n = p(1-p)^n w_n + (1-p)\sum_{j=1}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j b_j$$

For small p,

$$\begin{aligned}
w_{1,p} &= p, \\
w_{2,p} &\approx 3p^2, \\
w_{3,p} &\approx 13p^3, \\
w_{4,p} &\approx 85p^4 \\
a_n &= \sum_{j=0}^{n-1} \binom{n}{j} a_j
\end{aligned}$$

Is $n \mapsto w_{n,p}$ decreasing for some p?
Let us try to prove that $w_{n-1,p} - w_{n,p} \geq Cp(1-p)^{n-1}$ for some C.

$$\begin{aligned}
w_2 - w_3 &= 3p(1-p)^2(w_1 - w_2), \\
w_2 - w_3 &= -p(1-p)^2 w_2 + p^2(1-p)(1-w_1) + 2p(1-p)^2(w_1 - w_2) \\
w_1 - w_2 &= p(1-3p+2p^2) = p(1-2p)(1-p) = p(1-2p)(w_0 - w_1)
\end{aligned}$$

$$w_2 - w_3 = 3p(1-p)^2(w_1 - w_2)$$

For $n = 2, 3, ...,$

$$\begin{aligned}
w_{n-1,p} - w_{n,p} &= \sum_{j=0}^{n-2} \binom{n-1}{j} p^{n-1-j}(1-p)^j w_{j,p} - \sum_{j=0}^{n-1} \Big(\binom{n-1}{j} + \binom{n-1}{j-1} \Big) p^{n-j}(1-p)^j w_{j,p} \\
&= \Big(\sum_{j=0}^{n-2} \binom{n-1}{j} p^{n-1-j}(1-p)^{j+1} w_{j,p} \Big) - p(1-p)^{n-1} w_{n-1,p} \\
&\qquad - \sum_{j=0}^{n-2} \binom{n-1}{j} p^{n-j-1}(1-p)^{j+1} w_{j+1,p} \\
&= -p(1-p)^{n-1} w_{n-1,p} + \sum_{j=0}^{n-2} \binom{n-1}{j} p^{n-1-j}(1-p)^{j+1}(w_{j,p} - w_{j+1,p}) \\
&= -p^n + \sum_{k=0}^{n-2} p^{n-k}(1-p)^k w_k - p^{n-1-k}(1-p)^{k+1} w_{k+1} + \sum_{j=0}^{n-2} \binom{n-1}{j} p^{n-1-j}(1-p)^{j+1}(w_{j,p} - w_{j+1,p}) \\
&\geq -p(1-p)^{n-1} w_{n-1,p} + Cp \sum_{j=1}^{n-2} \binom{n-1}{j} p^{n-1-j}(1-p)^{2j} \\
&= -p(1-p)^{n-1} w_{n-1,p} + Cp(1-p+p^2)^{n-1} - Cp(1-p)^{2n-2}
\end{aligned}$$

so $w_{n-1,p} > w_{n,p}$ if $w_{n-1,p} < \frac{p^{n-2}}{(1-p)^{n-3}}$.

Let us study the latter, which holds for $p < 1/2$ and $n = 2, 3$. So, we have shown that for $p < 1/2$, we have $w_{1,p} > w_{2,p} > w_{3,p}$.

If $w_{k,p} \leq \frac{p^{k-1}}{(1-p)^{k-2}}$ for all $k \leq n-1$, then

$$\begin{aligned} w_{n,p} &= p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j w_{j,p} \\ &\leq p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j \frac{p^{j-1}}{(1-p)^{j-2}} \\ &= p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-1}(1-p)^2 \\ &= p^{n-1}(p + (1-p)^2(2^n - 2) \end{aligned}$$

so $w_{n-1,p} < \frac{p^{n-1}}{(1-p)^{n-2}}$ if $(1-p)^{n-2}(p + 2(1-p)^2(2^{n-1} - 1)) < 1$.

Furthermore,

$$w_{4,p} = p^4 + 4p^3(1-p)w_{1,p}$$

, which is certainly the case if $w_{n-2,p} < \frac{p^{n-2}}{(1-p)^{n-3}}$ or $w_k \leq \frac{p^k}{(1-p)^{k-1}}$ for all k.

$w_{1,p} > w_{2,p}$ if $w_{1,p} = p < 1-p$.

$w_{2,p} > w_{3,p}$ if $w_{2,p} < p = w_{1,p}$

Let us compute some bounds. Starting with n coins, we might in the first round set aside a single coin, and afterwards play optimally. This gives us

$$\begin{aligned} w_{n,p} &\le (1-p)^n + (1-p^n)w_{n-1,p}, \\ w_{n,p} - w_{n-1,p} &\le (1-p)^n - p^n w_{n-1,p}, \\ w_{n,p} &\le \sum_{k=0}^{n-1} (1-p)^{k+1} - p^{k+1} w_{k,p} = \frac{1-(1-p)^n}{p} - \sum_{k=0}^{n-1} p^{k+1} w_{k,p}. \end{aligned}$$

Applying the discrete Gronwall inequality gives

$$w_{n,p} \le \frac{1-(1-p)^n}{p} + \sum_{k=0}^{n-1} \frac{1-(1-p)^k}{p} p^{k+1} \exp\Big(- \sum_{j=k+1}^{n-1} p^{j+1} \Big)$$

Let us compute the first few values. Note that for $n = 1, 2$, there is literally no choice. For $n = 1$, set aside the head if possible. For $n = 2$, set aside one coin unless both come up heads, in which case you set aside both coins. For $n = 3$, if there are 0 or 1 coin, set aside one coin. If all are heads, set aside all three. However, if there are two heads, the first decision has to be made: With $w_{0,p} = 0$,

$$\begin{aligned} &w_{1,p} = 1-p, \\ &w_{2,p} = (2-p)(1-p)^2 + 2p(1-p)(1-p) = (2+p)(1-p)^2, \\ &w_{3,p} = (w_{2,p}+1)(1-p)^3 + 3p^2(1-p)(w_{1,p} \wedge w_{2,p}) + 3p(1-p)^2 w_{2,p}, \\ &w_{4,p} = (w_{3,p}+1)(1-p)^4 + 4p^3(1-p)(w_{1,p} \wedge w_{2,p} \wedge w_{3,p}) + 6p^2(1-p)^2(w_{2,p} \wedge w_{3,p}) + 4p(1-p)^3 w_{3,p}. \end{aligned}$$

Let us analyse $w_{3,p}$: We see that there is a distinction whether $w_{1,p} < w_{2,p}$, i.e. $1 < (2+p)(1-p) = 2 - p - p^2$, i.e. $p < \frac{\sqrt{5}+1}{2} \approx 0.618$. If $p < \frac{\sqrt{5}+1}{2}$, if two out of three coins come up heads, we set aside both of them, whereas for $p > \frac{\sqrt{5}+1}{2}$, we set aside only one.
Using the same approach, let us analyse $w_{4,p}$. Here, we need to understand when $w_{2,p} < w_{3,p}$. We distinguish two cases from above:
$p > \frac{\sqrt{5}+1}{2}$ (or $w_{2,p} < 1-p$). We already know $w_{1,p} > w_{2,p}$. Hence,

$$w_{3,p} = (1-p)^3 + (1-p^3)w_{2,p}.$$

From this, we see that $w_{3,p} < w_{2,p}$ iff $(1-p)^3 - p^3 w_{2,p} = (1-p)^2((1-p) - p^3(2+p)) < 0$, which is always true in this case and we have $w_{3,p} < w_{2,p} < w_{1,p}$.
$p < \frac{\sqrt{5}+1}{2}$ (or $w_{2,p} > 1-p$). Here, we get

$$w_{3,p} = (1-p)^3 + 3p^2(1-p)^2 + ((1-p)^3 + 3p(1-p))w_{2,p}.$$

Let us try to show $w_{n,p} \le w_{n-1,p}$ be induction. Assume this inequality holds for all k up to $n-1$. Then,

$$\begin{aligned} w_{n,p} &= (w_{n-1,p}+1)(1-p)^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j w_{n-1,p} \\ &= (1-p)^n + (1-p^n)w_{n-1,p} \ge (1-p)^n + (1-p^n)\frac{(1-p)^{n-1}}{p^{n-1}} = \frac{(1-p)^n}{p^n}\big(p^n + (p + p^2 + \cdots p^{n-1})\big), \end{aligned}$$

which is at most $w_{n-1,p}$ iff $(1-p)^n \le p^n w_{n-1,p}$.

Let us consider the strategy of always setting aside as much heads as possible. Here,

$$\begin{aligned} \mathbb{E}[Z_n|\mathcal{F}_{n-1}] &= (Z_{n-1}+1)(1-p)^n + \sum_{j=0}^{n-1} \binom{n}{j} p^{n-j}(1-p)^j Z_j \\ &= (1-p)^n + ((1-p)^n + np(1-p)^{n-1})Z_{n-1} + \sum_{j=0}^{n-2} \binom{n}{j} p^{n-j}(1-p)^j Z_j. \end{aligned}$$

Hence, with $a_{n,n} = 1$ for all n,

$$\mathbb{E}\Big[\sum_{k=0}^{n} a_{kn} Z_k | \mathcal{F}_{n-1}\Big] = (1-p)^n + (a_{n-1,n} + ((1-p)^n + np(1-p)^{n-1}))Z_{n-1} + \sum_{k=0}^{n-2}(a_{kn} + \binom{n}{k} p^{n-k}(1-p)^k)Z_k$$
$$\stackrel{!}{=} \sum_{k=0}^{n-1} a_{k,n-1} Z_k.$$

We find

$$1 = a_{n-1,n} + ((1-p)^n + np(1-p)^{n-1}),$$
$$a_{k,n-1} = a_{kn} + \binom{n}{k} p^{n-k}(1-p)^k, \qquad k = 0, ..., n-2$$

Let $J \sim B(n, 1-p)$. Then,

$$Z_n = 1_{J=n}(1 + Z_{n-1}) + 1_{J<n} Z_J = Z_J - 1_{J=n}(Z_n - Z_{n-1} - 1) = Z_J$$

So, with $Z_0 = 0$,

$$\mathbb{E}[Z_1] = (1-p),$$
$$\mathbb{E}[Z_2] = (1-p)^2(1+1-p) + 2p(1-p)^2 = (1-p)^2 + (1-p)^2(1+p) = (1-p)^2 + (1-p)(1-p^2),$$
$$\mathbb{E}[Z_3] = (1-p)^3 + ((1-p)^3 + 3p(1-p)^2)((1-p)^2 + (1-p)(1-p^2)) + 3(1-p)p^2(1-p)$$

Specifically, the test statistic, hereafter referred to as Delta Likelihood, represents the difference in the likelihood of the ancestry vector $\widehat{q}^K$ estimated for a given genotype if drawn from any combination of the allele frequencies (p) represented among the $K = 5$ reference groups minus the likelihood of the ancestry vector $\widehat{q}^L$ estimated strictly from allele frequencies among the $L = 4$ domestic pig reference groups. Accordingly, if p_{km} is the frequency of the reference allele at locus m in the kth pig reference group, and $X_m = \{0, 1, 2\}$ is the number of reference allele the individual under consideration carries, the Delta Likelihood statistic for this individual can be calculated as:

$$\Delta\ell(p|X) := \frac{1}{2M} \sum_{m=1}^{M} X_m \log\Big(\frac{\langle \widehat{q}^K, p_{\cdot m}\rangle}{\langle \widehat{q}^L, p_{\cdot m}\rangle}\Big) + (2 - X_m)\log\Big(\frac{1 - \langle \widehat{q}^K, p_{\cdot m}\rangle}{1 - \langle \widehat{q}^L, p_{\cdot m}\rangle}\Big).$$

Here, $\langle q, p_{\cdot m}\rangle := \sum_{k=1}^{K} q_k p_{km}$ is the probability of obtaining the reference allele for an individual with admixture q at locus m. Note that we assume that all loci are bi-allelic; see Pfaffelhuber & Rohde (2023) for a deeper analysis of this likelihood and methods to compute $\widehat{q}^K$ and $\widehat{q}^L$.

For a bi-allelic marker with frequency p_k in reference group k, one can compute $\bar{p} := \frac{1}{K}(p_1 + \cdots + p_K)$ as well as $\overline{p(1-p)} := \frac{1}{K}(p_1(1-p_1) + \cdots + p_K(1-p_K))$. Then, F_{ST} is given by

$$F_{ST} = \frac{\bar{p}(1-\bar{p}) - \overline{p(1-p)}}{\bar{p}(1-\bar{p})}$$

for this marker. In general, high F_{ST} means that allele frequencies in the groups differ.